Indus Weave Textiles
Why overtime cost keeps rising

PeopleOps Data Hackathon • Solo • Python + SQL + SQLite
6 months: 1 Apr – 30 Sep 2026 • ~1,200 employees • Headcount fell, OT bill rose

The question: is this more work, or more paid-but-not-worked overtime?

1 • Paid OT is not worked OT

Computed OT from biometric punches: 29,116 hours in 6 months
Approved / paid OT: 36,402 hours
Gap: 7,286 hours

Apr gap: 856 hours → Sep gap: 1,748 hours
OT bill: PKR 2.81m (Apr) → PKR 3.48m (Sep), +23.5%
Worked OT was flat: 4,670 – 5,051 hours per month.

Action: pay OT only from computed OT. No computed OT, no payment.

2 • Approvals pay for days with no OT worked

3,518 of 19,666 approved OT requests (17.9%)
had zero computed OT on that day.

584 were Present days with zero OT after shift end.
The rest were Absent, Missing Punch, On Leave, Holiday or Weekly Off days.

Faisalabad Plant is the worst: 14,418 paid vs 9,895 computed hours.

Action: block approval when computed OT = 0, and audit Faisalabad first.

3 • Pay did not stop on separation

5 employees were paid after separation.
Total: PKR 1,983,188 across 21 employee-months.

Example: Umair Baloch, separated 23 Apr 2026,
paid May–Sep: PKR 682,100.

Only 1 employee-month had pay with zero Present days in that month —
the same control failure, not a workload story.

Action: payroll must stop automatically on separation_date, with a monthly Q8 check.

How we proved it — and what to fix next

- Raw loaded untouched, all TEXT, with _row_hash — 472,907 rows
- DEV-SITE-01 clock was 60 min fast from 1 Jul 2026 — corrected, documented
- 20,465 Missing Punch days (9.4%) — fix biometric check-out discipline
- 20 employees already exceed full-year Annual leave in 6 months
- DQ: 22/22 checks pass, raw = staging + rejected

Next: computed-OT approval gate, separation payroll stop, SITE device audit.

Demo • What to look at

Run: python run_pipeline.py
Then: python analytics/run_queries.py

Q1–Q10: analytics/q01..q10.sql → analytics/results/*.csv
Warehouse: indus_weave.db • dim_employee (SCD2), dim_department, dim_shift, dim_date
Facts: fact_attendance_daily, fact_leave_day, fact_overtime_approved, fact_payroll_line

Questions? I can explain every line — start with Q5, Q6, Q8.